

PACKETS OF ENERGY

In 1900, Max Planck showed a way to avoid the ultraviolet catastrophe (see p.27) and make the theoretical emissions of black bodies (see p.26) match with their measured behavior. What if energy was being released not in a continuous stream, but as small, discrete bursts (or packets of energy), each with a distinct wavelength? Planck called these bursts “light quanta,” and assumed that their production had something to do with the emission process rather than being a property of light itself (see p.14).



Classical physics

In the pre-quantum view, properties of particles, such as the energy they hold, vary continuously and may have any value.